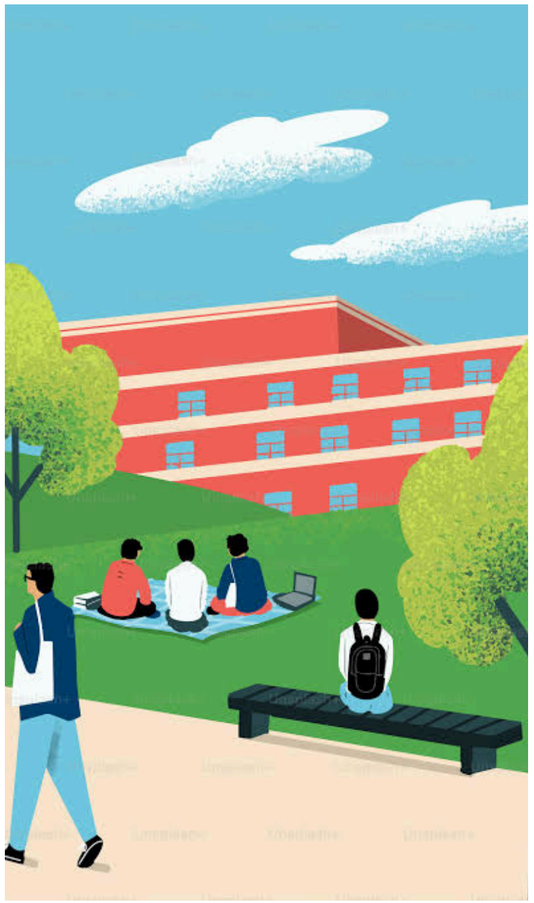


Capstone 2: Smart Lost & Found System

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Introduction

Our project focuses on developing a Smart Campus Lost & Found System that helps students report, search for, and recover lost items within the university.

The system provides a centralized platform where users can submit lost and found item reports, search for matching items, and receive notifications about potential matches.

The main goal is to make the process of finding lost belongings faster, easier, and more organized.

This presentation covers the project plan, system requirements, Use Case Modeling, BPMN, WBS, and the project timeline.

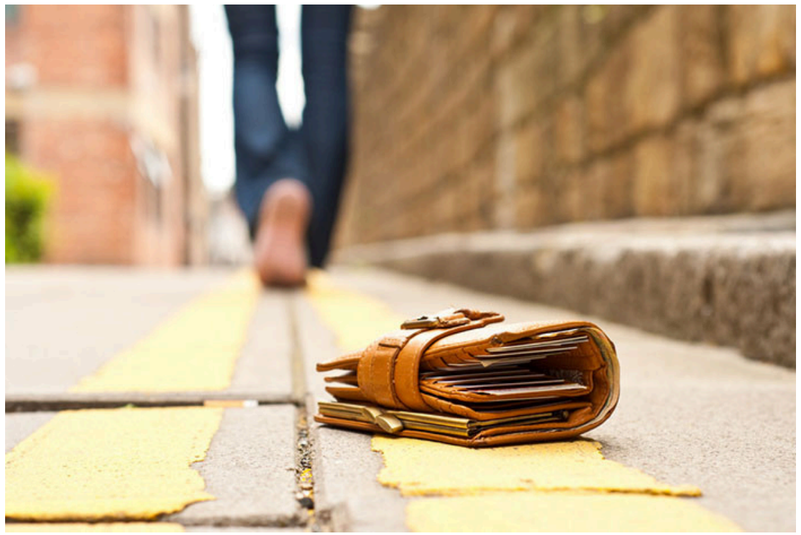




Project Plan







Project Background

Students frequently lose personal belongings on campus, while finding the owners of discovered items can be difficult. A centralized digital system can help solve this problem.

Project Objectives

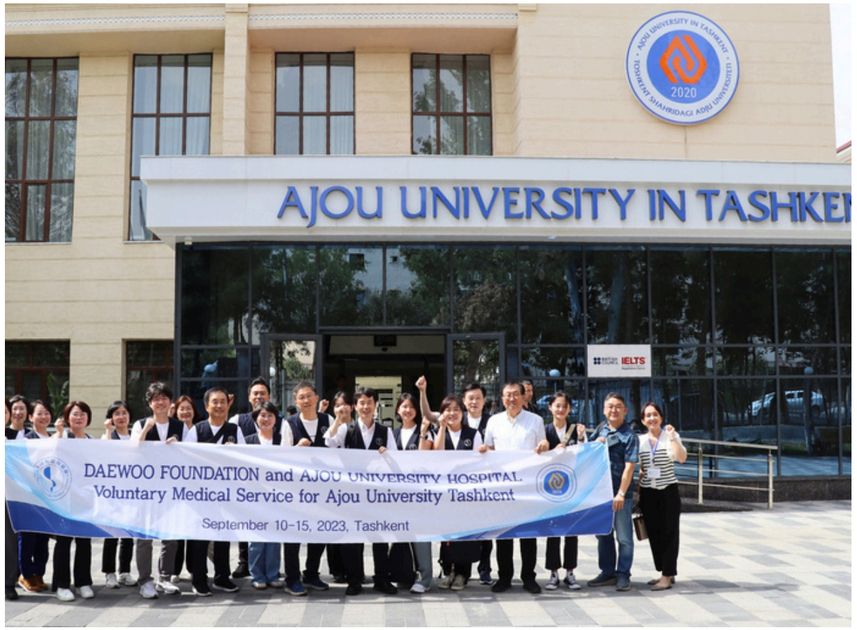
- Provide a platform for reporting lost and found items.
- Enable users to search for matching items.
- Notify users about potential matches.
- Verify ownership before returning items.
- Improve the efficiency of campus lost-and-found procedures.



Project Plan

Target Audience

- University students
- Item finders
- Campus administrators

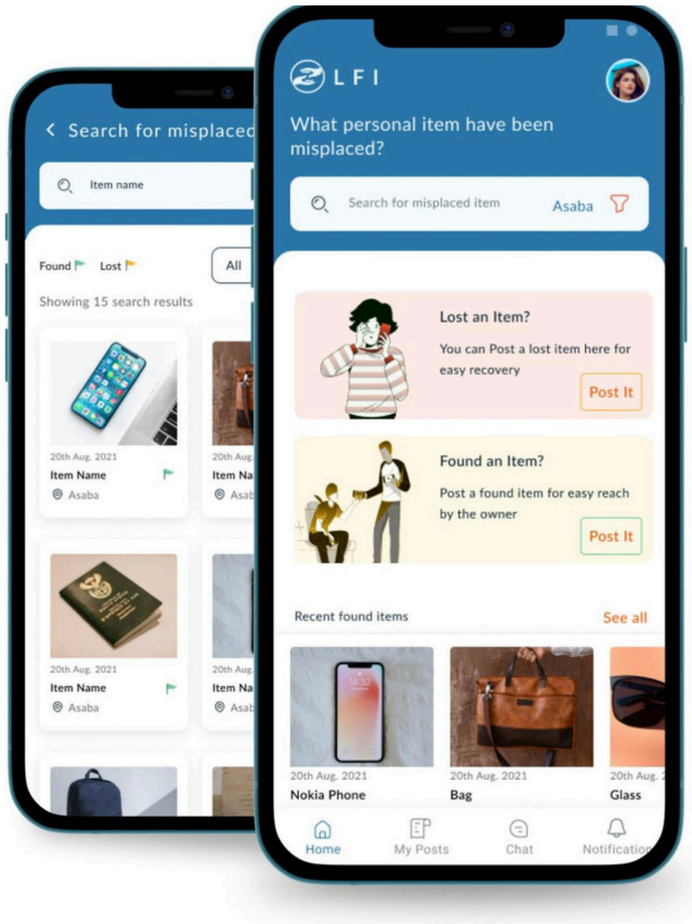


Project Scope

The project includes user registration, item reporting, searching, notifications, claim verification, and item return tracking.







System Requirements



01

Functional Requirements

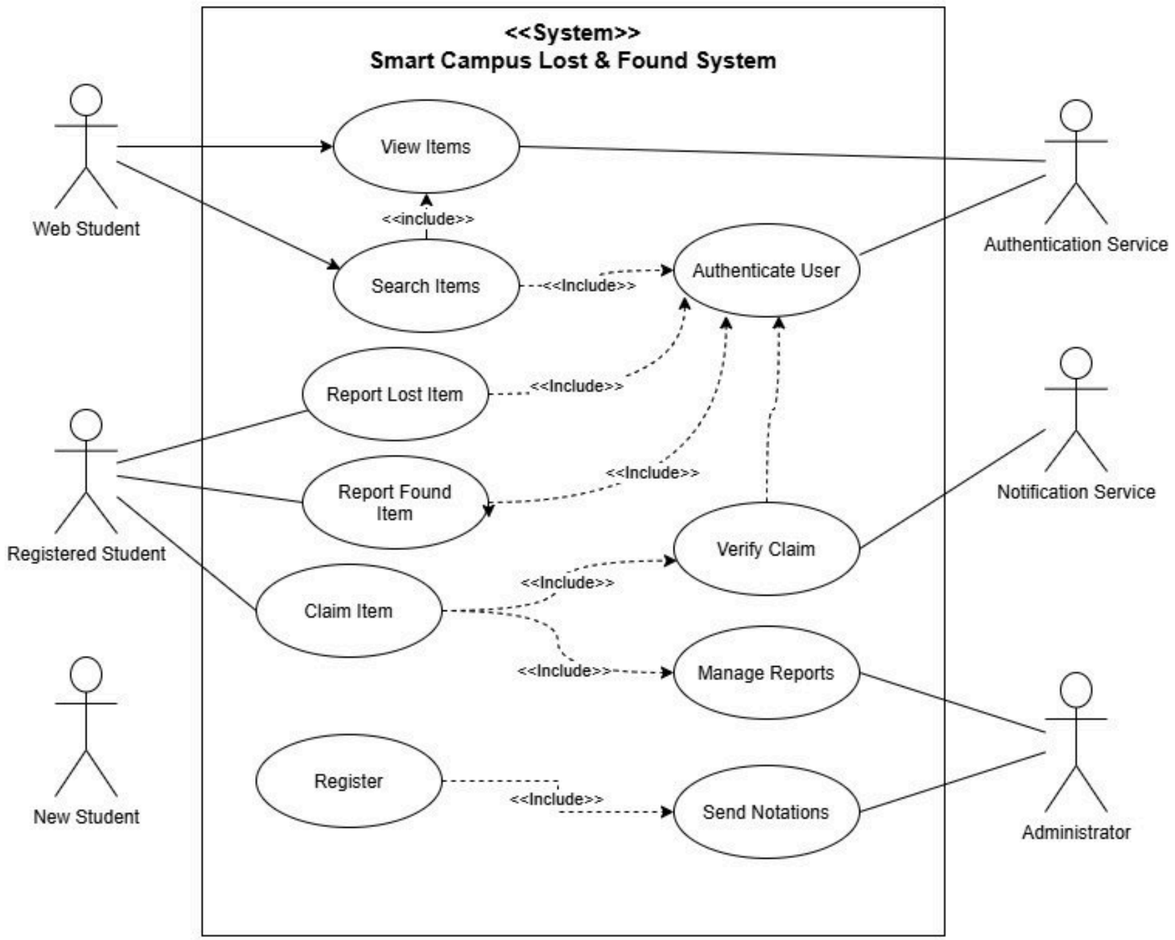
- User registration and authentication.
- Reporting lost and found items.
- Searching and matching item descriptions.
- Sending notifications about potential matches.
- Submitting and verifying ownership claims.
- Updating item status after successful return.
- Managing reports through an administrator interface

02

Non-Functional Requirements:

- **Usability:** The system should be easy to navigate.
- **Security:** User accounts and personal information should be protected.
- **Reliability:** Item reports should be stored accurately.
- **Performance:** Search results should be provided efficiently.
- **Accessibility:** The platform should be convenient to access for campus users

Use Case Modeling



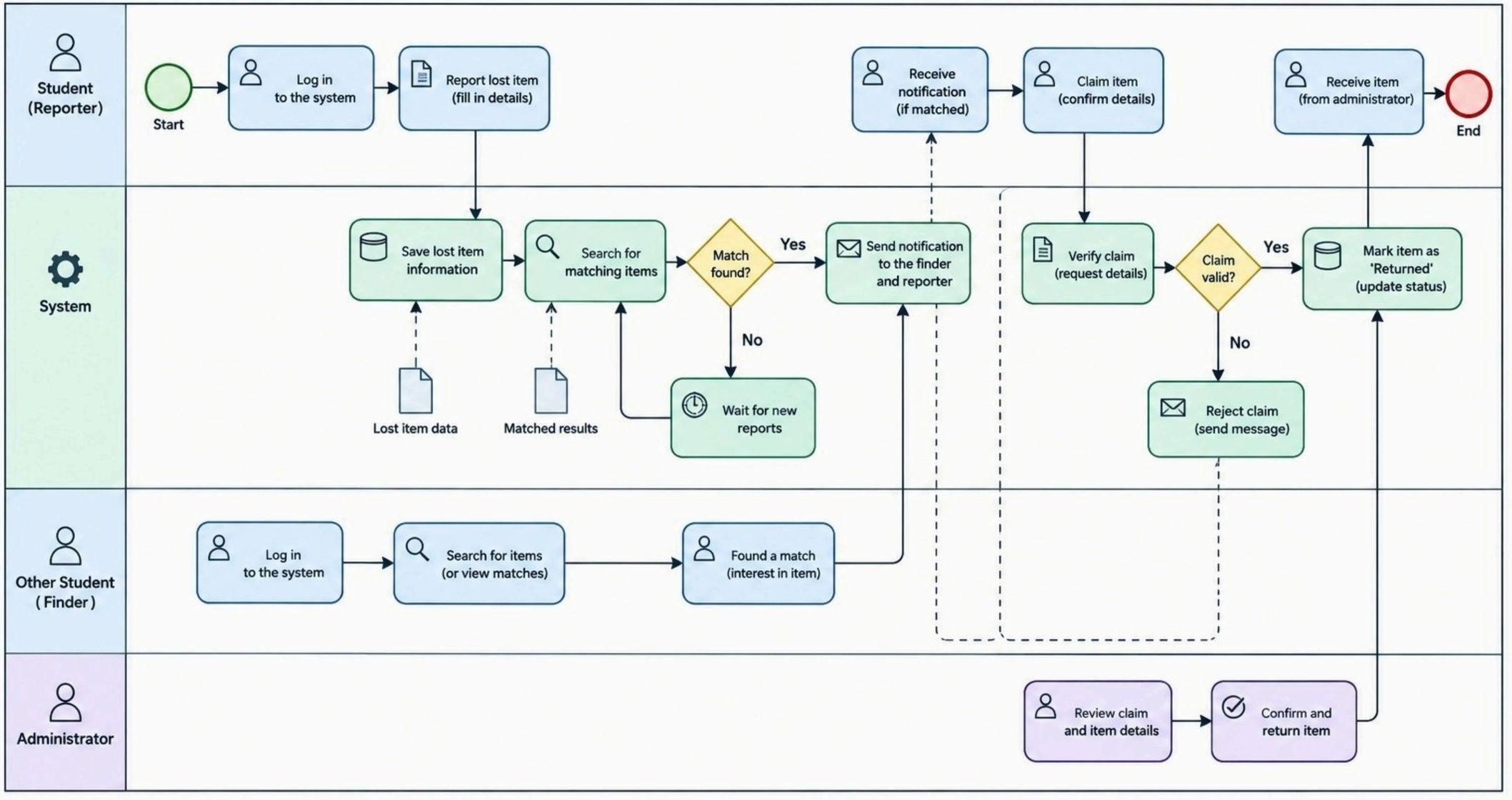
Use Case Modeling illustrates how different users interact with the Smart Campus Lost & Found System.

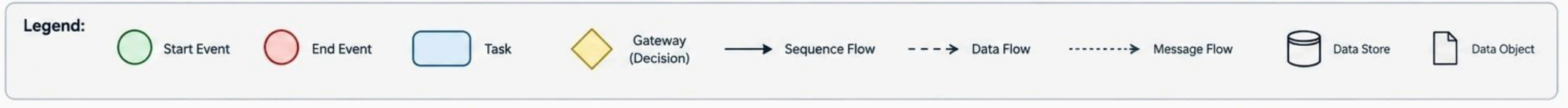
Main Actors:

- New Student: Registers to access the system.
- Web Student: Views and searches for items.
- Registered Student: Reports lost or found items and submits claims.
- Authentication Service: Supports user authentication.
- Notification Service: Supports notifications about potential matches and claims.
- Administrator: Manages reports and supports the verification process.

The Use Case Diagram defines the main system functions and the interactions between users and external services.

Smart Campus Lost & Found System – BPMN 2.0





WBS (Work Breakdown Structure)

Smart Campus Lost & Found System

Development of a system for finding and returning lost items on campus.

1.0 Smart Campus Lost & Found System

A web-based system to help students find and return lost items.

Project Goal

Create a convenient and reliable system where students can report lost items, find matching items, and return them to their owners.

1.1 User Management

Authentication, roles and user profiles.

- 1.1.1 User registration and login
- 1.1.2 Role management
- 1.1.3 User profiles
- 1.1.4 Access control

1.2 Lost Item Management

Reporting, storing and managing lost items.

- 1.2.1 Report lost item
- 1.2.2 Save item information
- 1.2.3 View and update item details
- 1.2.4 Track item status

1.3 Matching & Search

Find matching items and handle results.

- 1.3.1 Search for items
- 1.3.2 Match lost and found items
- 1.3.3 Process matches
- 1.3.4 Wait for new reports

1.4 Notifications & Claims

Inform users and process claim requests.

- 1.4.1 Send notifications
- 1.4.2 Notify finder and reporter
- 1.4.3 Submit claim request
- 1.4.4 Confirm claim details

1.5 Verification & Return

Validate claims and return items to owners.

- 1.5.1 Verify claim
- 1.5.2 Admin review
- 1.5.3 Approve or reject claim
- 1.5.4 Transfer item to owner
- 1.5.5 Mark item as returned



Smart Campus Lost & Found System

Project Gantt Chart

Development of a system for finding and returning lost items on campus.

Project Duration
12 weeks

WBS	Task / Work Package	Duration
1.1	**User Management**	**2 weeks**
1.1.1	User registration and login	3 days
1.1.2	Role management	3 days
1.1.3	User profiles	2 days
1.1.4	Access control	2 days
1.2	**Lost Item Management**	**2 weeks**
1.2.1	Report lost item	5 days
1.2.2	Save item information	3 days
1.2.3	View and update item details	5 days
1.2.4	Track item status	2 days
1.3	**Matching & Search**	**2 weeks**
1.3.1	Search for items	5 days
1.3.2	Match lost and found items	5 days
1.3.3	Process matches	3 days
1.3.4	Wait for new reports	Ongoing
1.4	**Notifications & Claims**	**2 weeks**
1.4.1	Send notifications	3 days
1.4.2	Notify finder and reporter	3 days
1.4.3	Submit claim request	3 days
1.4.4	Confirm claim details	3 days
1.5	**Verification & Return**	**2 weeks**
1.5.1	Verify claim	4 days
1.5.2	Admin review	4 days
1.5.3	Approve or reject claim	3 days
1.5.4	Transfer item to owner	3 days
1.5.5	Mark item as returned	2 days



1.1 User Management · 1.2 Lost Item Management · 1.3 Matching & Search · 1.4 Notifications & Claims · 1.5 Verification & Return

Timeline shows planned tasks and dependencies.



Conclusion

The Smart Campus Lost & Found System aims to simplify the process of finding and returning lost belongings within the university. It provides a centralized platform where students can report lost and found items, search for potential matches, and receive notifications.

Expected Outcomes:

- Faster identification of lost items through an organized search and matching process.
- Improved communication between students and finders through notifications.
- More reliable ownership verification before returning items.
- Reduced time and effort required to locate lost belongings

Conclusion:

The project plan defines the system's objectives, scope, and requirements. The Use Case Diagram illustrates user interactions, while BPMN describes the main business process. The WBS organizes project tasks, and the Gantt Chart outlines the 12-week development schedule.



Thank You

